

Figure 5.4 (a) Same as figure 5.3a, except when one observer moves toward the other, her *now* slice rotates into the future, not the past, of the other observer. (b) Same as 5.3b—a greater separation yields a greater deviation in conceptions of *now,* for the same relative velocity— with the rotation being toward the future instead of the past.

While surprising, none of this generates any contradiction or paradox because, as we explained above, the farther away something is, the longer it takes to receive light it emits and hence to determine that it belongs on a particular now-list. For instance, even though John Wilkes Booth's approaching the State Box at Ford's Theatre will belong on Chewie's new now-list if he gets up and walks away from earth at about 9.3 miles per hour,[2] he can take no action to save President Lincoln. At such an enormous distance, it takes an enormous amount of time for messages to be received and exchanged, so only Chewie's descendants, billions of years later, will actually receive the light from that fateful night in Washington. The point, though, is that when his descendants use this information to update the vast collection of past now-lists, they will find that the Lincoln assassination belongs on the same now-list that contains Chewie's just getting up and walking away from earth. And yet, they will also find that a moment before Chewie got up, his now-list contained, among many other things, you, in earth's twenty-first century, sitting still, reading these words.[3]

Similarly, there are things about our future, such as who will win the U.S. presidential election in the year 2100, that seem completely open: more than likely, the candidates for that election haven't even been born, much less decided to run for office. But if Chewie gets up from his chair and walks toward earth at about 6.4 miles per hour, his now-slice—his conception of what exists, his conception of what has happened—*will* include the selection of the first president of the twenty-second century. Something that seems completely undecided to us is something that, for